

Deliverable 3: Predictive Model Development and Performance Evaluation

MAI643 Project | Team 6: Stroke Prediction

1. Project Title and Team Members

Project Title: Machine Learning-Driven Risk Stratification for Stroke Using Clinical and Lifestyle Factors

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2. Revisions from Previous Deliverable

No changes were made to the raw dataset itself. The main revision in Deliverable 3 is methodological: all preprocessing operations were embedded directly inside cross-validation and GridSearchCV pipelines so that imputation, encoding, scaling, SMOTE, and PCA are learned only from the training folds. This prevents data leakage.

In addition, the report has been expanded to include every generated figure together with an explicit interpretation, in response to the feedback that all visual outputs should be discussed in the documentation.

3. Model Evaluation Metrics

The dataset is strongly imbalanced: 249 stroke cases out of 5110 observations (4.87%). Because of this, a model can achieve very high accuracy by predicting the majority class only. For that reason, the evaluation used a broader set of metrics.

- Accuracy: retained for completeness and comparison with prior studies.
- Precision: shows how reliable positive stroke predictions are.
- Recall (Sensitivity): measures how many true stroke cases are detected.
- F1-score (weighted): summarizes predictive performance while accounting for imbalance.
- ROC-AUC: evaluates how well the model ranks stroke vs non-stroke cases across thresholds.
- Balanced Accuracy: used as the primary model-selection criterion because it gives equal weight to both classes.

Balanced accuracy and recall were chosen as primary selection criteria because raw accuracy is misleading on this dataset (~95% accuracy is achievable by predicting no-stroke for every patient). Balanced accuracy penalises both false negatives and false positives equally, while recall directly measures the fraction of actual stroke cases detected — the clinically critical quantity.

4. Initial Model Experimentation

- Dataset size after dropping the identifier: 5110 rows and 10 predictive features.

- Train/test split: 4088 training rows and 1022 test rows.
- Cross-validation protocol: 5-fold stratified cross-validation.
- Candidate classifiers: Logistic Regression, KNN, Random Forest, SVM, AdaBoost.
- Preprocessing variants: V1 baseline, V2 scaling, V3 scaling + SMOTE, V4 scaling + SMOTE + PCA.

Initial cross-validation results by model

Model	Accuracy	Balanced Accuracy	Precision	Recall	Weighted F1	ROC_AUC
LogisticRegression	0.8466	0.6291	0.0665	0.3882	0.8713	0.8375
AdaBoost	0.8321	0.6197	0.0599	0.3845	0.8609	0.8189
SVM	0.8680	0.5862	0.0605	0.2739	0.8851	0.7068
KNN	0.8795	0.5575	0.1214	0.2008	0.8929	0.6547
RandomForest	0.9308	0.5190	0.1842	0.0627	0.9199	0.7931

Initial cross-validation results by dataset version

Version	Accuracy	Balanced Accuracy	Precision	Recall	Weighted F1	ROC_AUC
V4	0.7830	0.6742	0.1200	0.5537	0.8387	0.7717
V3	0.8018	0.6497	0.1122	0.4813	0.8500	0.7731
V1	0.9503	0.5028	0.0790	0.0071	0.9277	0.7660
V2	0.9504	0.5024	0.0827	0.0060	0.9277	0.7380

The initial experiments showed a clear split between accuracy-oriented and sensitivity-oriented behaviour. Versions V1 and V2 gave very high accuracy, but they almost completely failed to identify stroke cases. Versions V3 and V4 lowered raw accuracy but substantially improved recall and balanced accuracy.

5. Figure-by-Figure Analysis of Initial Experiments

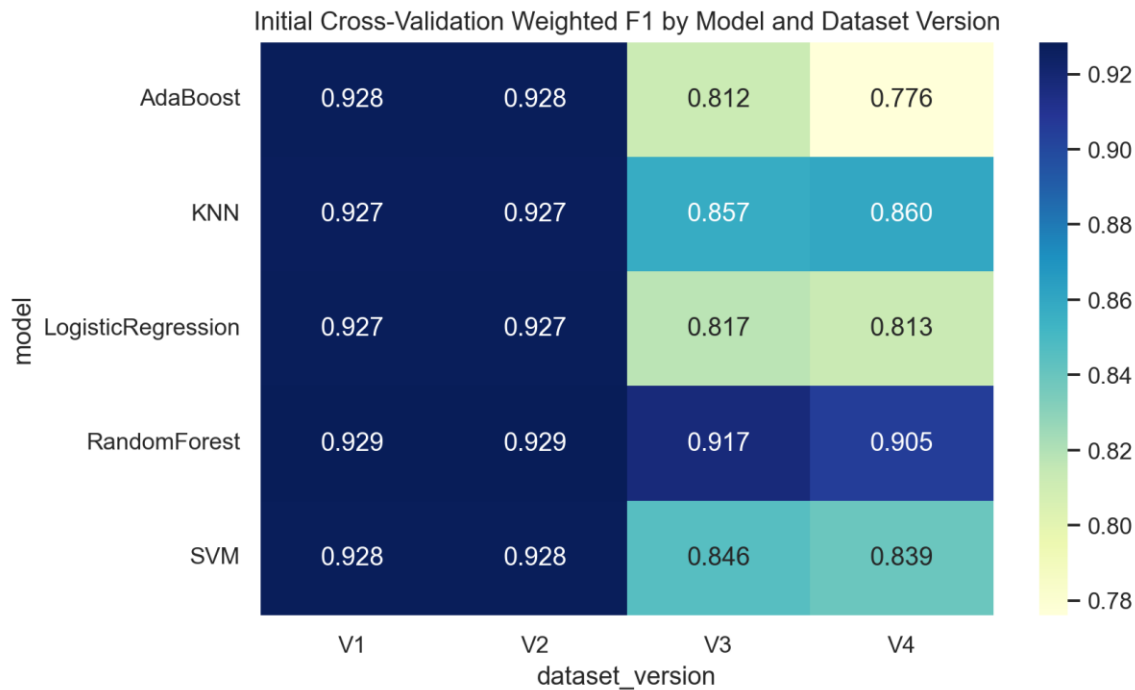


Figure 1. Initial cross-validation weighted F1-score heatmap.

This heatmap shows that V1 and V2 appear strong when evaluated by weighted F1-score because the majority class dominates the metric. Random Forest and KNN look competitive here, but this is misleading: the figure reflects overall classification quality more than minority-class detection. This is why weighted F1-score alone was not used for final selection.

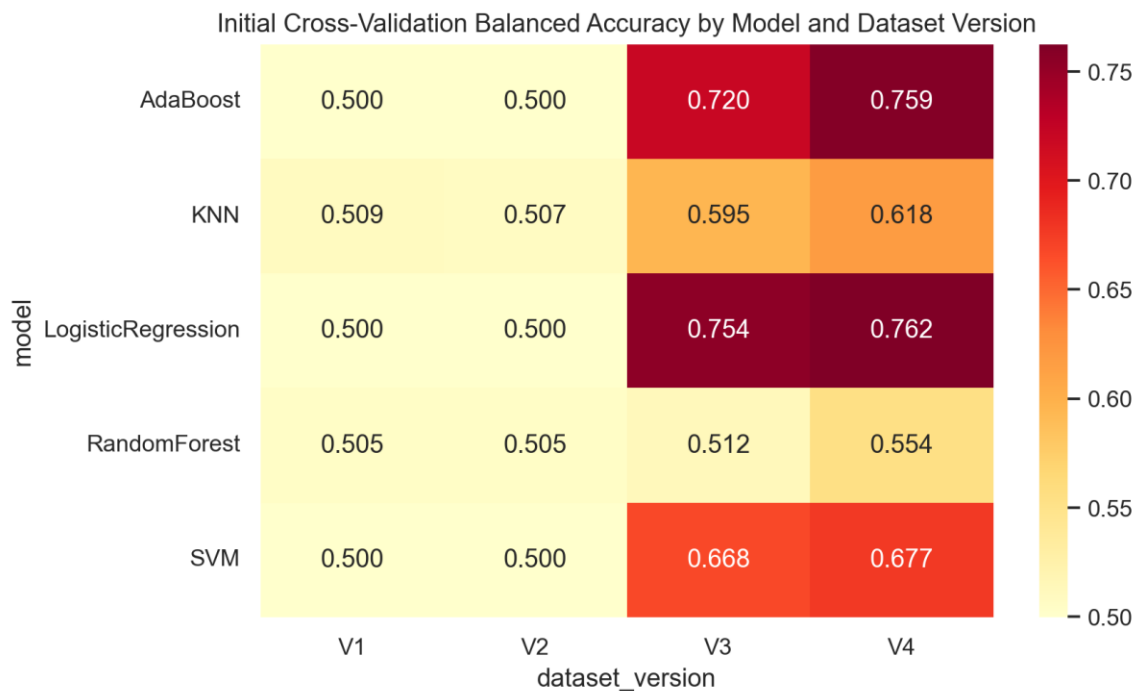


Figure 2. Initial cross-validation balanced-accuracy heatmap.

This figure is more important for the project goals because balanced accuracy weights stroke and non-stroke equally. The best combinations shift to V3 and V4, especially Logistic Regression and AdaBoost. This confirms that SMOTE-based preprocessing is necessary to obtain useful clinical sensitivity rather than only majority-class accuracy.

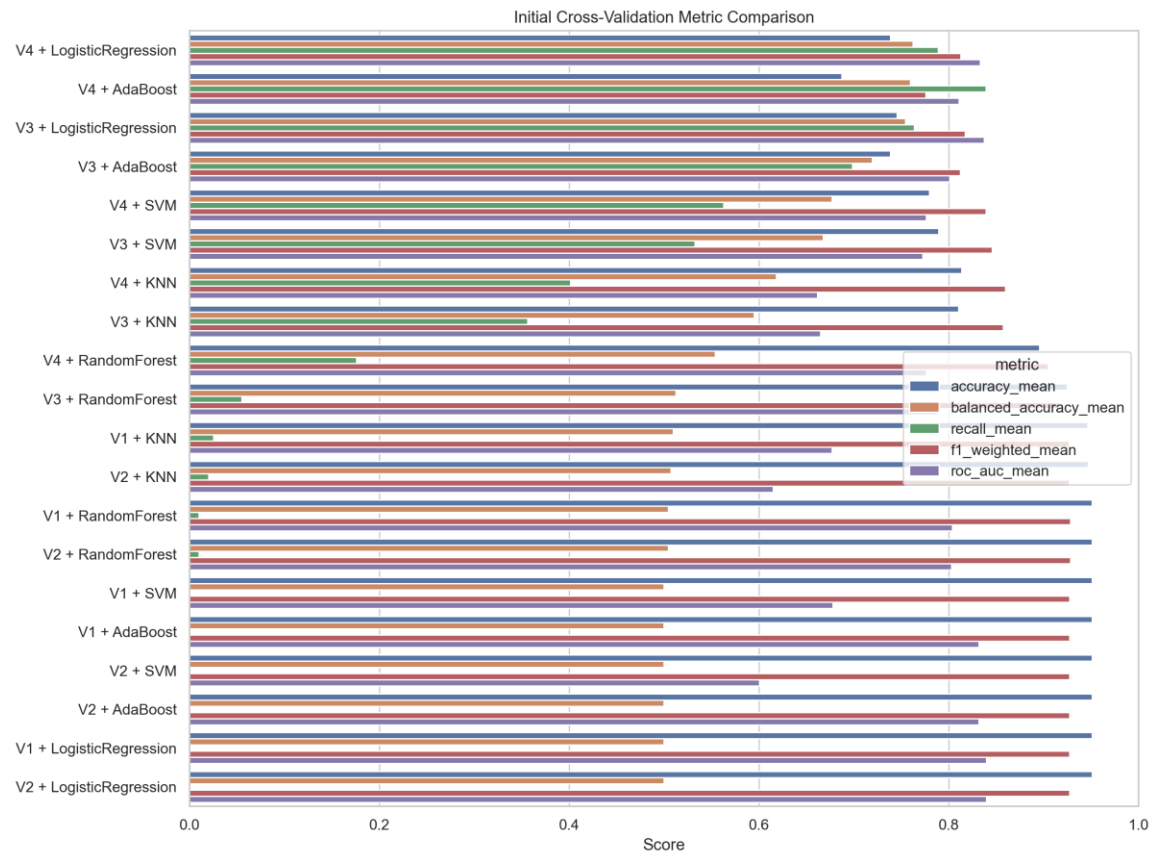


Figure 3. Initial cross-validation metric comparison across all model-version combinations.

This figure makes the trade-offs visible in one place. V1 and V2 combinations have high accuracy and often respectable ROC-AUC, but recall is near zero. In contrast, V3 and V4 combinations sacrifice some accuracy while raising recall and balanced accuracy. The pattern also shows that AdaBoost and Logistic Regression are the most suitable finalists for a recall-sensitive screening task.

These model+version combinations achieve a high CV ROC-AUC (good probabilistic ranking) yet recall = 0.0 at the default 0.5 threshold (always predict no-stroke). This confirms that SMOTE is necessary to shift the decision boundary toward the minority class; AUC alone is insufficient to detect strokes without rebalancing.

Combinations with high ROC-AUC but zero recall at the default threshold

Version	Model	ROC_AUC	Recall	Balanced Accuracy
V1	AdaBoost	0.8318	0.0000	0.5000
V2	AdaBoost	0.8318	0.0000	0.5000
V1	LogisticRegression	0.8397	0.0000	0.4999
V2	LogisticRegression	0.8397	0.0000	0.4999

6. Selection of Best-Performing Models and Dataset Versions

- Selected models: LogisticRegression, AdaBoost
- Selected dataset versions: V4, V3

Selection was based primarily on balanced accuracy and recall, not raw accuracy. This is appropriate for a stroke-screening problem because missing a stroke case is much more serious than generating additional follow-up checks.

7. Pipeline Definition

The final pipelines integrated preprocessing and modeling into a single reproducible workflow.

Pipeline for V3:

1. Median imputation for numerical features
2. Standard scaling for numerical features
3. Most-frequent imputation for categorical features
4. One-hot encoding for categorical features
5. SMOTE applied inside the training fold
6. Classifier (Logistic Regression or AdaBoost)

Pipeline for V4:

7. Same preprocessing as V3
8. PCA retaining 95% explained variance
9. Classifier (Logistic Regression or AdaBoost)

8. Hyperparameter Tuning

GridSearchCV with 5-fold stratified cross-validation was applied to the four finalist combinations: V4 + Logistic Regression, V4 + AdaBoost, V3 + Logistic Regression, and V3 + AdaBoost.

- Logistic Regression grid: solver in ["liblinear", "lbfgs"], C in [0.1, 1.0, 5.0, 10.0], class_weight in [None, "balanced"]
- AdaBoost grid: n_estimators in [50, 100, 200], learning_rate in [0.5, 1.0, 1.5]

9. Final Model Evaluation

Held-out test results of tuned finalist models

Combination	Accuracy	Balanced Accuracy	Precision	Recall	Weighted F1	ROC_AUC
V4 LogisticRegression	0.7524	0.7750	0.1413	0.8000	0.8222	0.8444
V3 LogisticRegression	0.7515	0.7745	0.1408	0.8000	0.8215	0.8443
V3 AdaBoost	0.7025	0.7677	0.1243	0.8400	0.7871	0.8337
V4 AdaBoost	0.7299	0.7632	0.1307	0.8000	0.8066	0.8292

The best final model was V4_LogisticRegression with balanced accuracy 0.7750, recall 0.8000, precision 0.1413, weighted F1 0.8222, and ROC-AUC 0.8444.

Of 50 actual stroke cases in the held-out test set, 40 were correctly flagged (TP) and 10 were missed (FN — false negatives). 243 non-stroke patients were incorrectly flagged as high-risk (FP — false positives). In a population screening context, minimising false negatives is the primary goal: a missed stroke (FN) carries a far higher clinical cost than an unnecessary follow-up (FP).

Precision of 0.141 is low by design. The model flags 243 false positives alongside 40 true positives because it was optimised for recall (sensitivity) rather than precision. In a population-level screening tool this is the correct tradeoff: the goal is to minimise missed strokes and refer borderline cases for clinical follow-up, not to achieve high precision at the cost of undetected events.

10. Figure-by-Figure Analysis of Final Evaluation

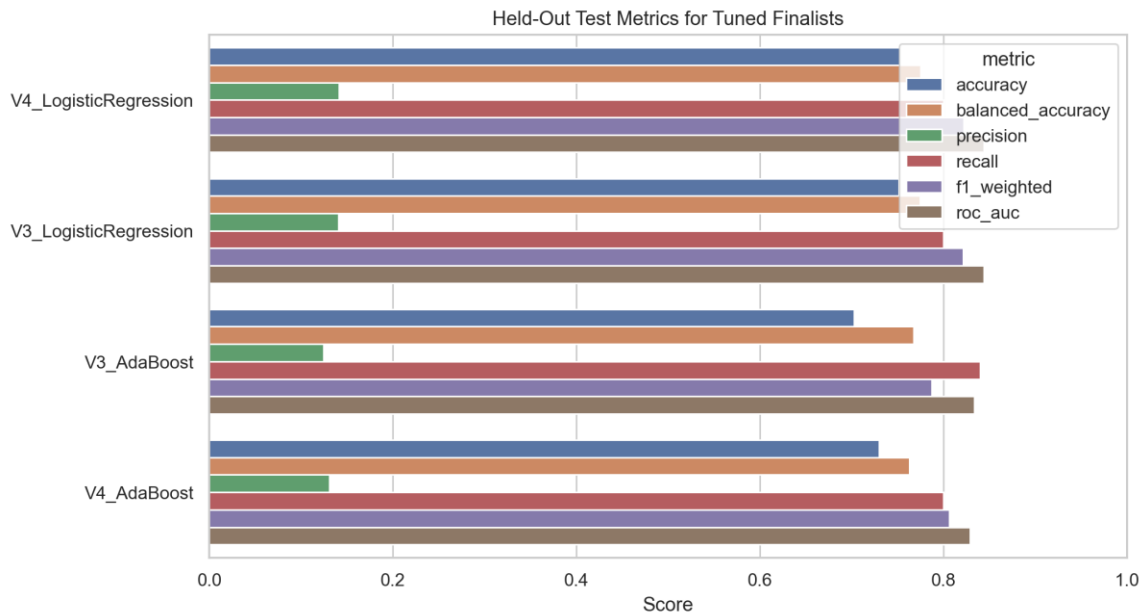


Figure 4. Held-out test metric comparison for the tuned finalist models.

This figure compares the four tuned finalists on the unseen test split. V4 + Logistic Regression and V3 + Logistic Regression are the most balanced combinations overall, with nearly identical recall and ROC-AUC but slightly stronger balanced accuracy and weighted F1 for V4. AdaBoost reaches the highest recall in V3, but at the cost of lower precision, lower weighted F1, and more false positives.

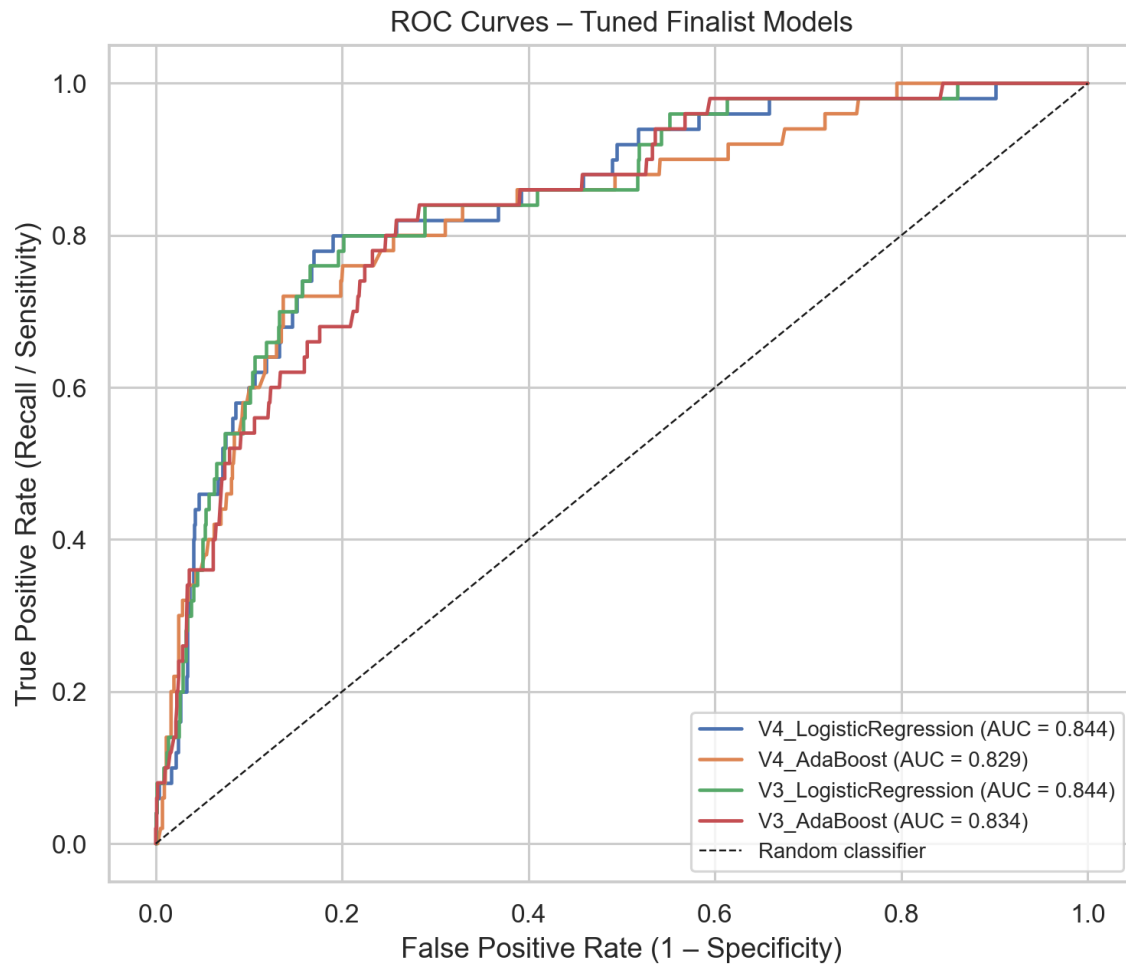


Figure 5. ROC curves for the tuned finalist models.

The ROC curves show that all four finalists separate stroke and non-stroke cases reasonably well, with AUC values around 0.83 to 0.84. The curves are close to each other, which indicates that the main difference between these models is not ranking ability but threshold behaviour and error trade-off. This supports the decision to use balanced accuracy and recall, rather than AUC alone, for final selection.

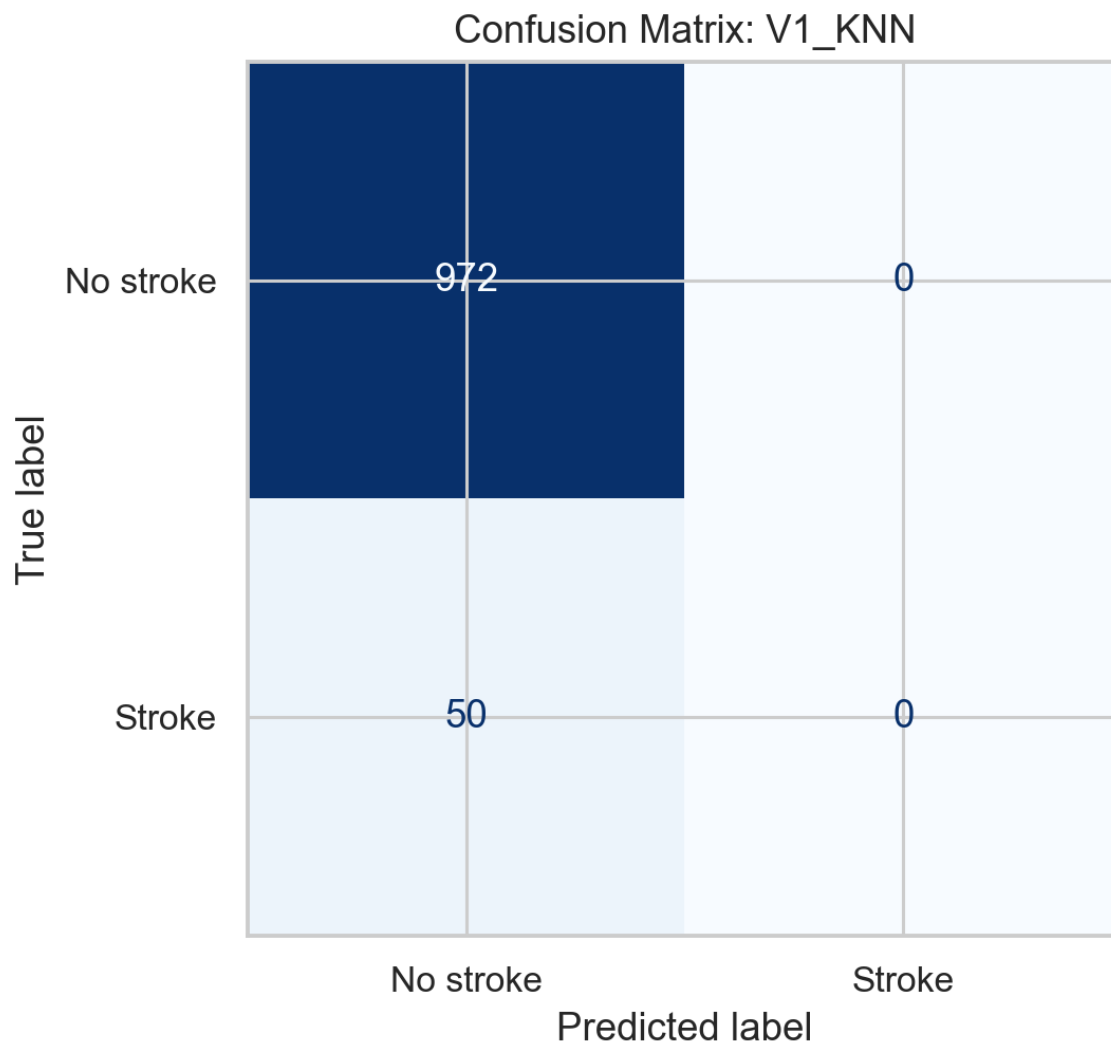


Figure 6. Confusion matrix for V1 + KNN.

This baseline model achieves high overall accuracy mainly by predicting the majority class. The matrix shows that only a very small number of stroke cases are detected, which makes this pipeline unsuitable despite its strong accuracy score.

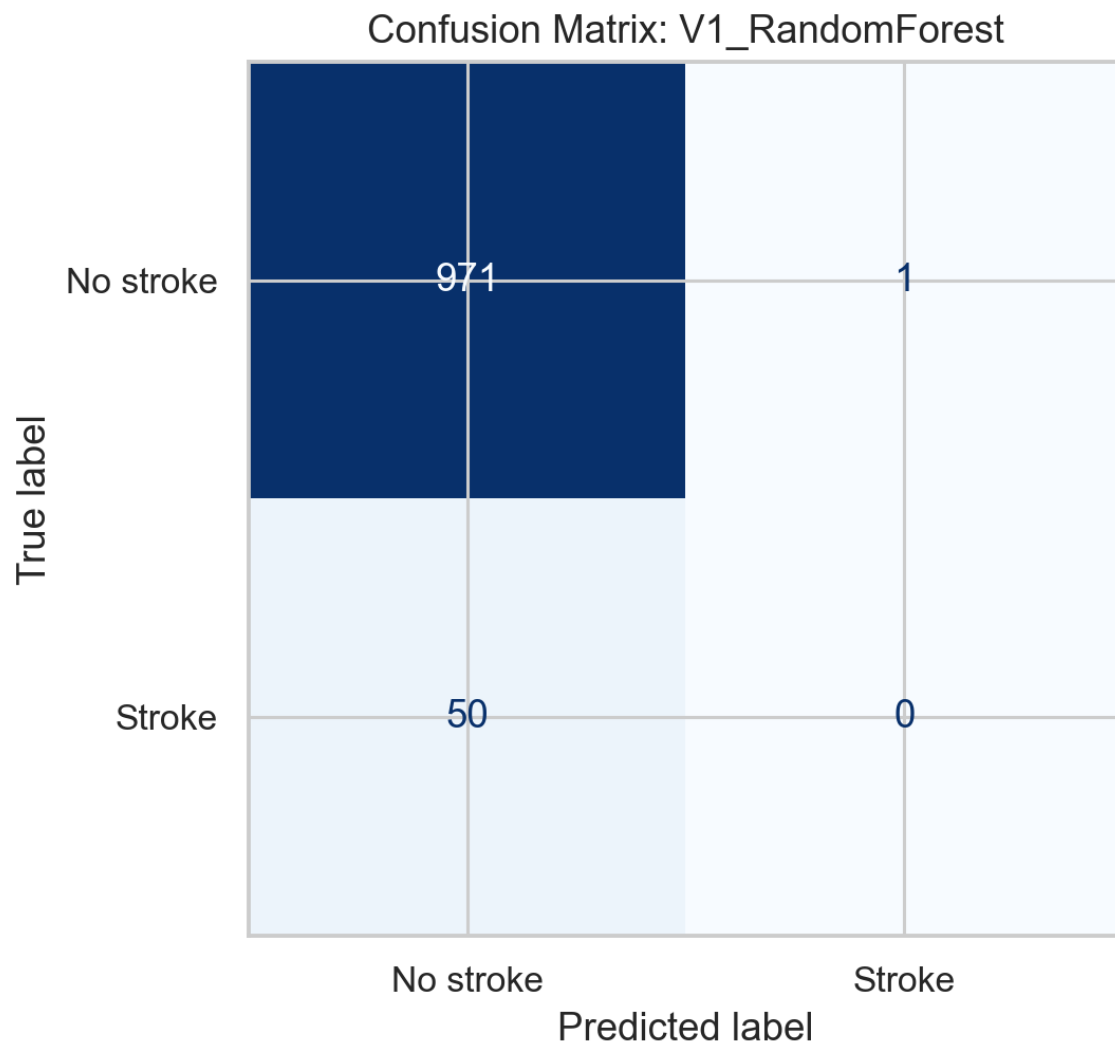


Figure 7. Confusion matrix for V1 + Random Forest.

The V1 Random Forest behaves similarly to V1 KNN but even more strongly favors the majority class. The result confirms that without rebalancing, tree-based models can look strong numerically while still failing the core objective of stroke detection.

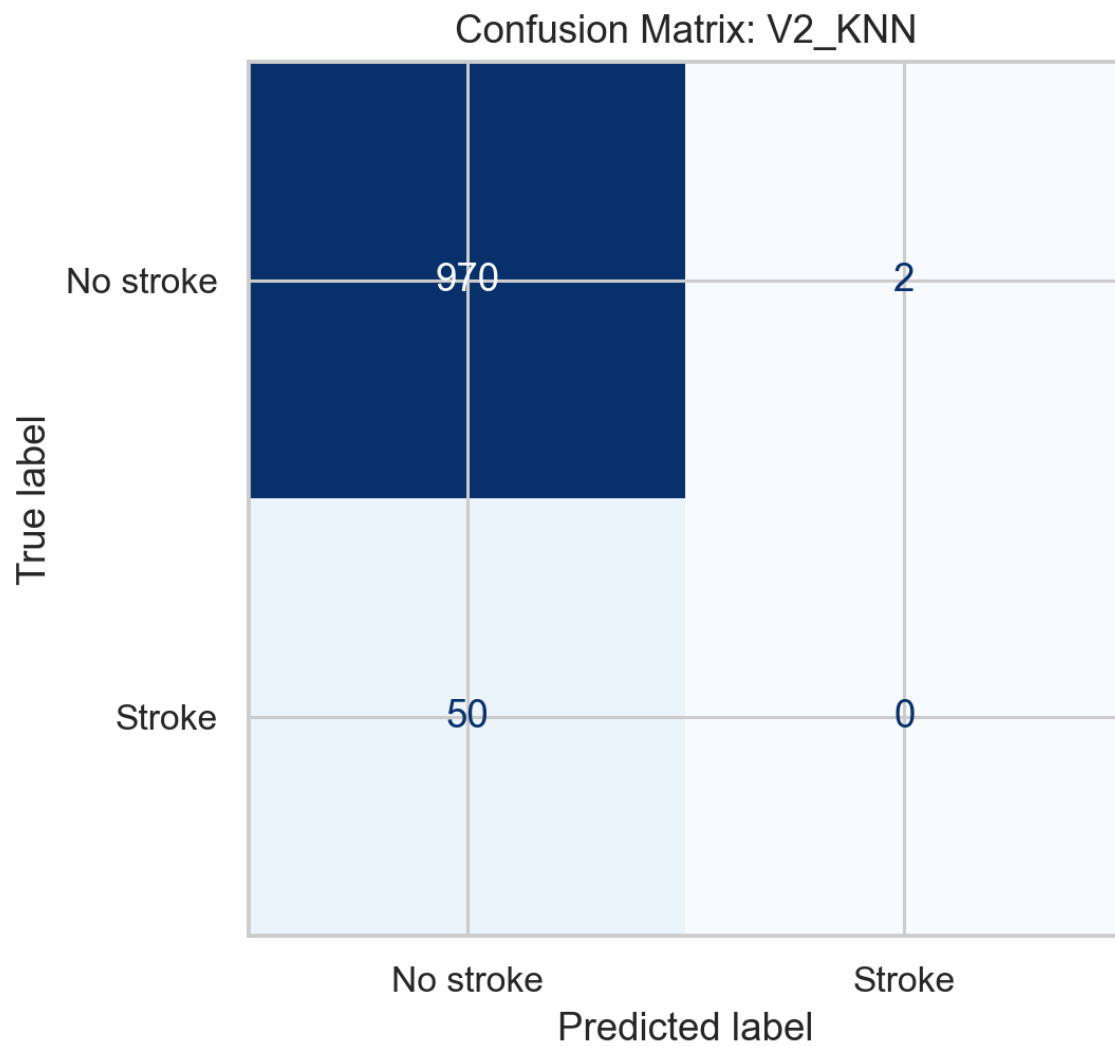


Figure 8. Confusion matrix for V2 + KNN.

Adding scaling in V2 does not solve the class-imbalance problem for KNN. The figure still shows near-total collapse to non-stroke predictions, meaning scaling alone is not enough without resampling.

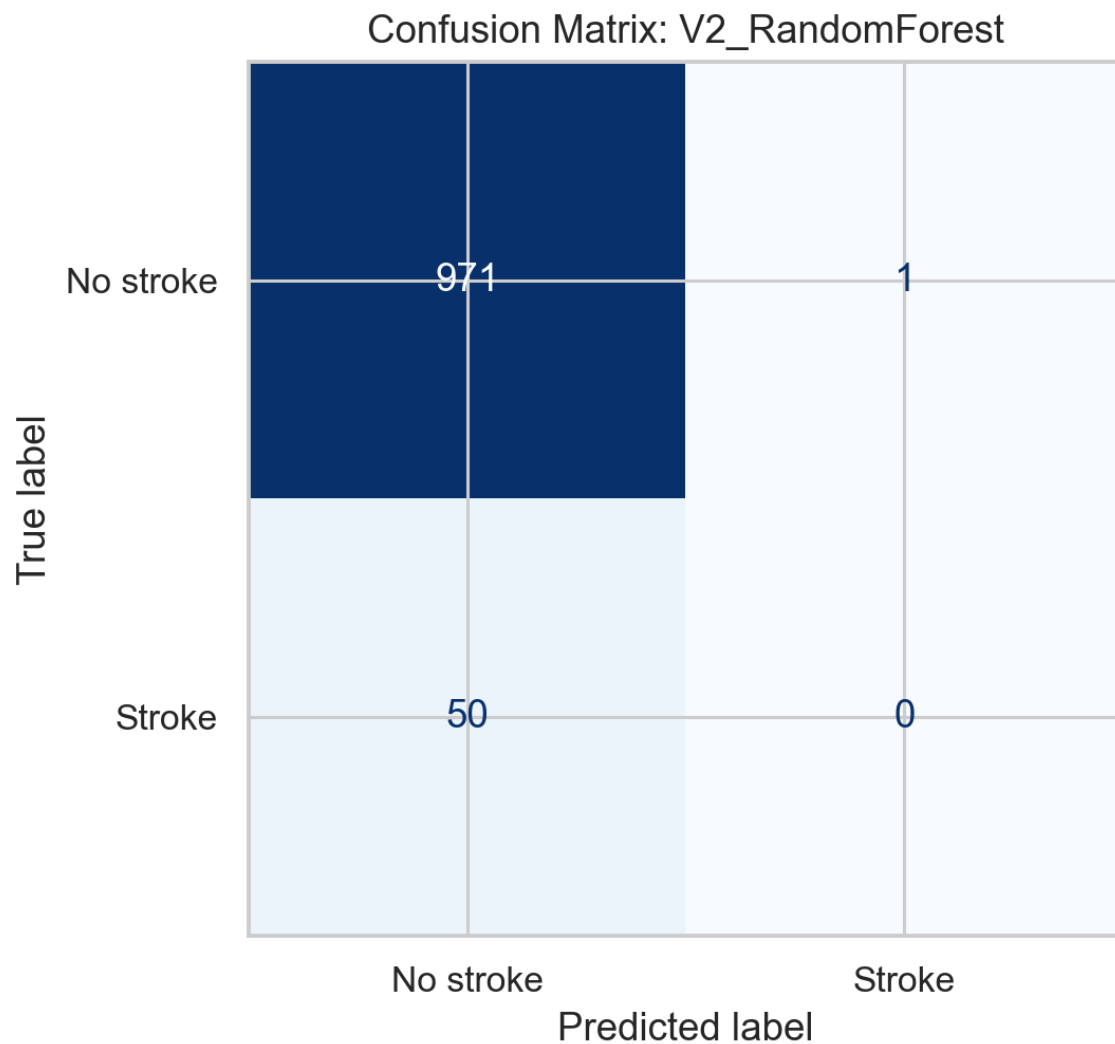


Figure 9. Confusion matrix for V2 + Random Forest.

Random Forest remains dominated by the majority class even after scaling is introduced in V2. This supports the conclusion from the heatmaps: preprocessing changes that do not rebalance the class distribution fail to recover minority-class recall.

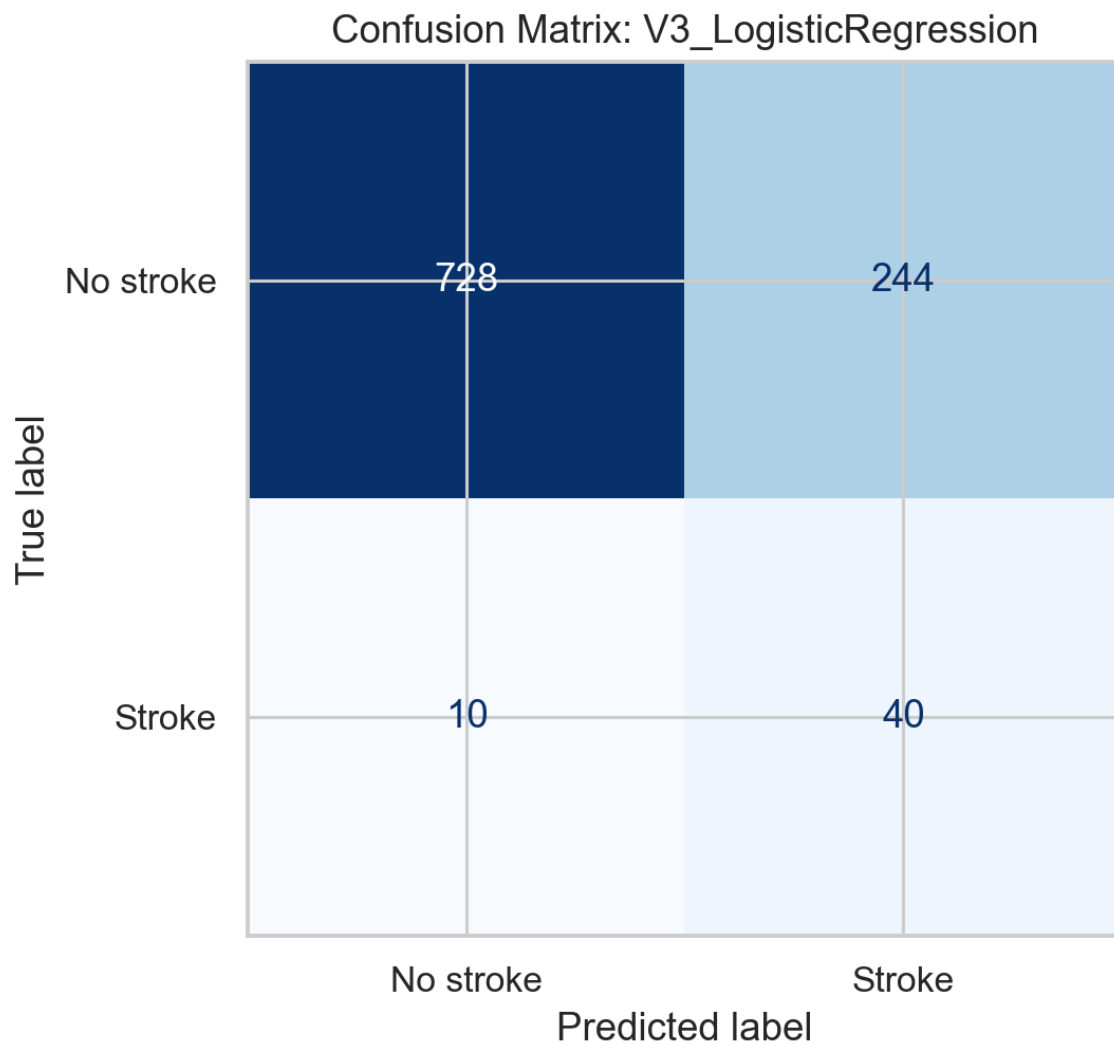


Figure 10. Confusion matrix for V3 + Logistic Regression.

Once SMOTE is introduced, Logistic Regression shifts decisively toward stroke detection. The model captures 40 of 50 stroke cases on the held-out test set, but generates many false positives. This is still a better clinical trade-off than missing most stroke cases.

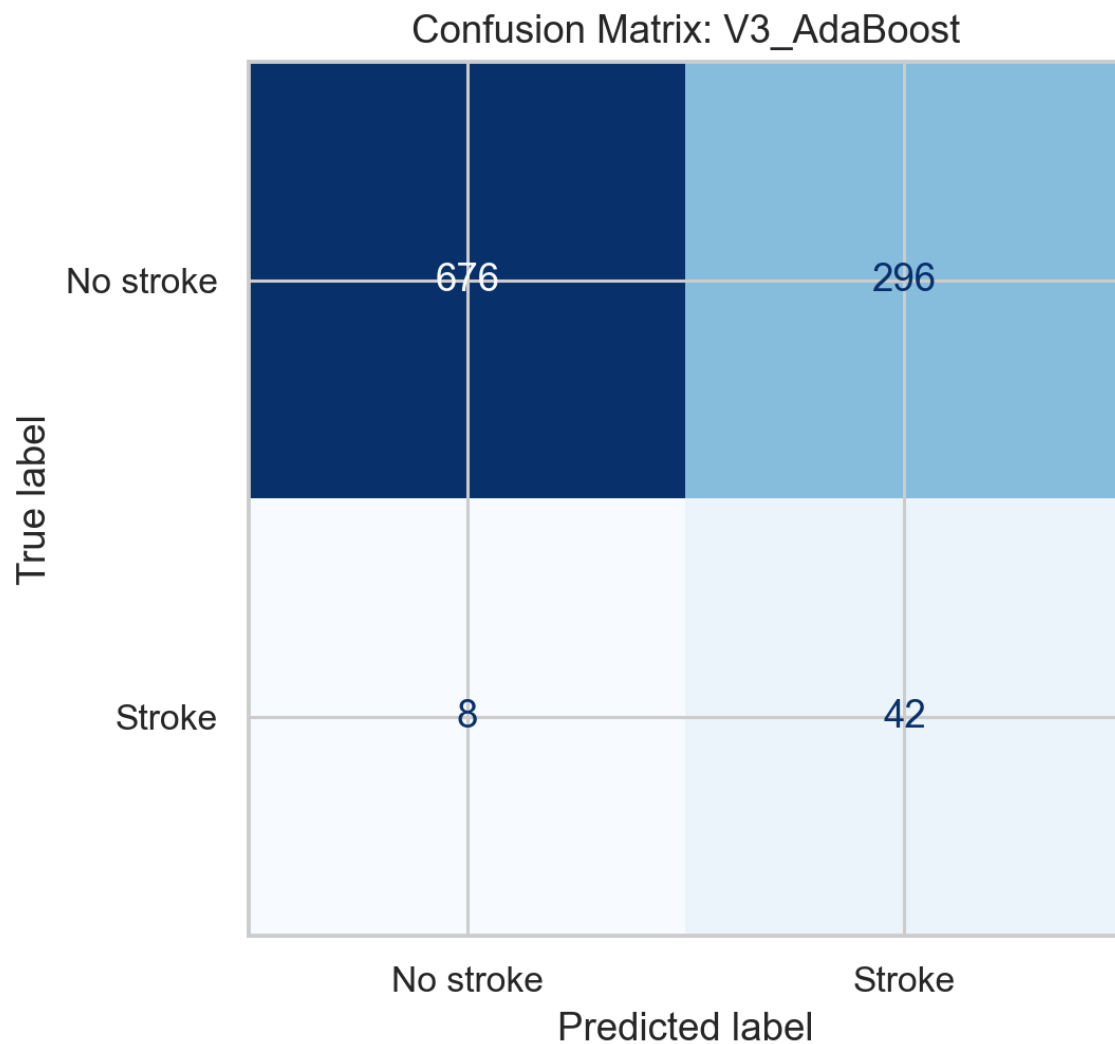


Figure 11. Confusion matrix for V3 + AdaBoost.

This model has the highest recall among the finalists, correctly identifying 42 of the 50 stroke cases. However, it also produces the largest number of false positives, which reduces precision and lowers overall balanced performance relative to the best Logistic Regression models.

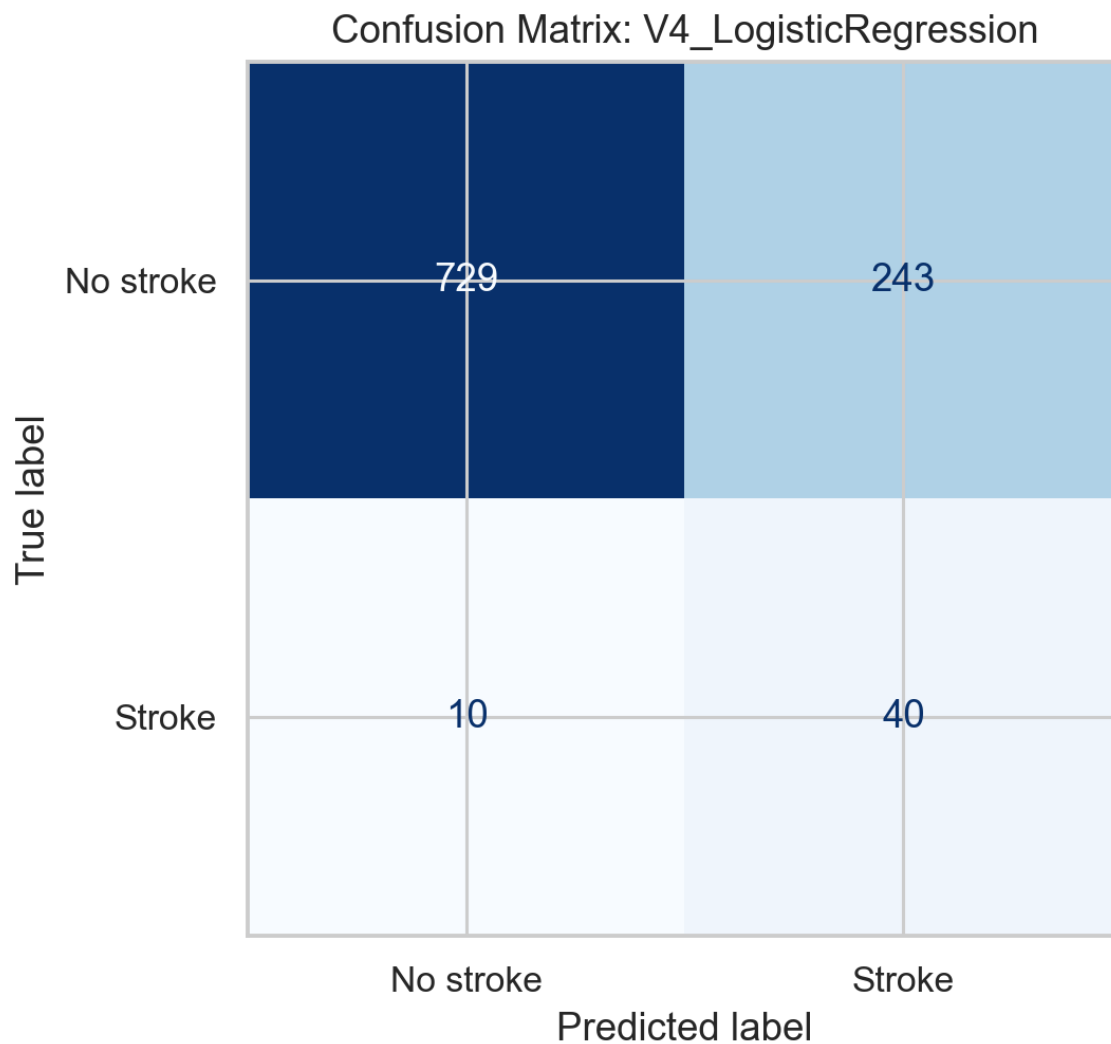


Figure 12. Confusion matrix for V4 + Logistic Regression.

This is the best final model. It preserves high recall while slightly reducing false positives compared with the other finalists, which gives it the best overall balance of metrics. The result also suggests that PCA did not harm predictive quality and may have helped stabilize generalization.

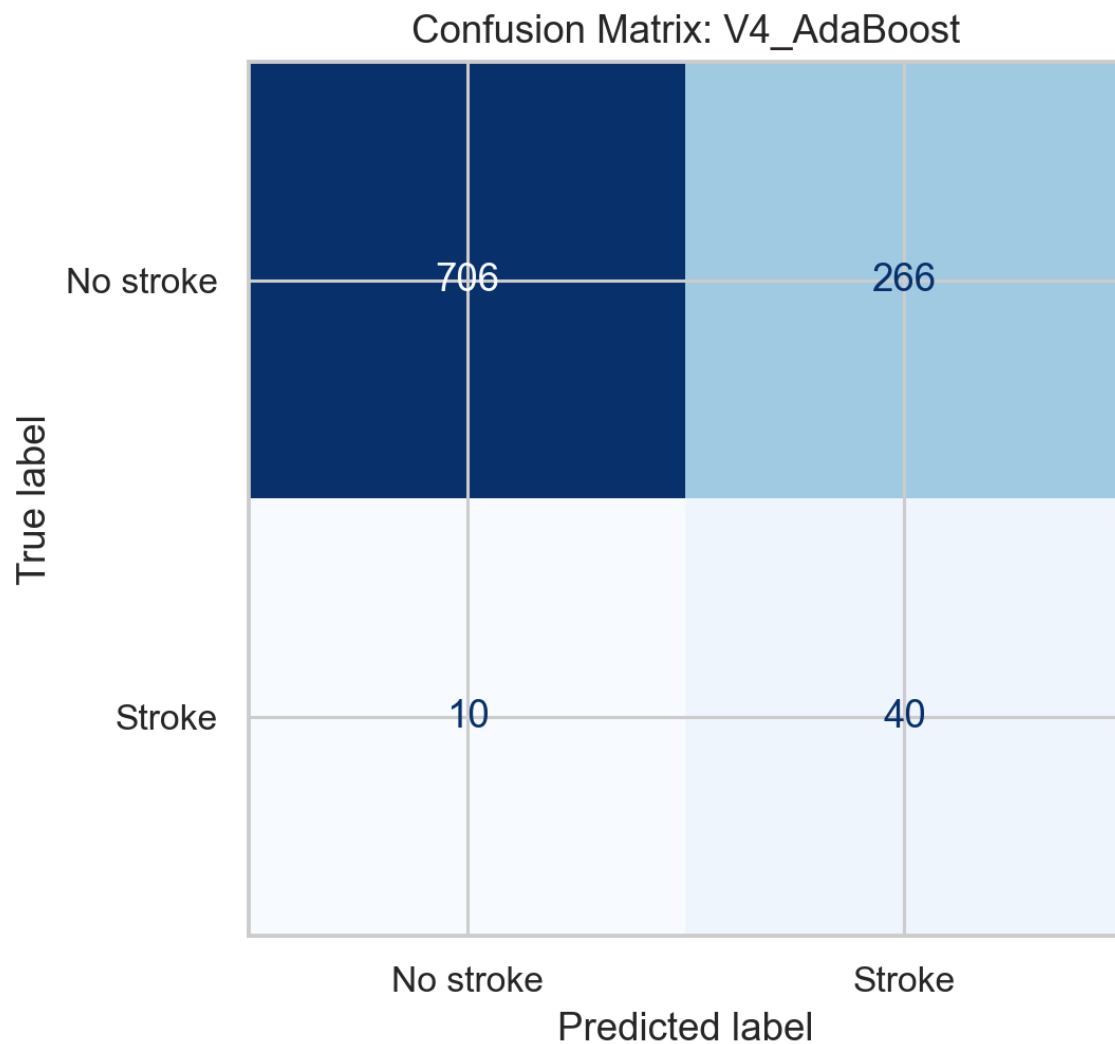


Figure 13. Confusion matrix for V4 + AdaBoost.

The V4 AdaBoost model remains competitive, but it does not improve enough over V3 AdaBoost to become the best choice. Its confusion matrix shows a similar sensitivity profile to the best model, but with a less favorable trade-off between correct detections and false alarms.

11. Discussion

Key Findings

1. Raw accuracy was not a reliable evaluation metric on this dataset. Near-95% accuracy was achievable while detecting almost no stroke cases.
2. SMOTE-based versions V3 and V4 improved recall and balanced accuracy substantially, which made them more appropriate for the medical objective.
3. Logistic Regression was the most stable final model. It preserved high recall while achieving the strongest overall metric balance.
4. PCA in V4 did not degrade performance and slightly improved the final Logistic Regression result compared with V3.
5. ROC-AUC confirmed good ranking performance across several models, but the confusion matrices showed that AUC alone is insufficient if the default classification threshold collapses to the majority class.

Comparison with Literature

Study	Dataset/setup	Best model	Reported result	Comparison
Dritsas and Trigka (2022)	Same Kaggle stroke dataset; balanced setting with ensemble methods	Stacking ensemble	Precision 0.974, Recall 0.974, F-measure 0.974, Accuracy 0.98	These results are much higher than ours, but the evaluation setting differs because their work is based on a balanced dataset rather than the original imbalanced distribution.
Rakshit and Shrestha (2021)	Kaggle stroke dataset; accuracy-focused comparison	Decision Tree	Accuracy 100%	This result is far above ours and likely reflects different split methodology or leakage-sensitive evaluation. Our experiments show that accuracy alone is not a trustworthy comparison on this dataset.
This work	Same Kaggle stroke dataset; preprocessing inside CV/GridSearchCV; held-out imbalanced test split	V4 + Logistic Regression	Balanced Accuracy 0.775, Recall 0.80, Precision 0.141, ROC-AUC 0.844	Lower raw accuracy, but the evaluation is stricter and explicitly optimized for clinically relevant detection rather than majority-class performance.

Domain-Specific Validation

The findings are consistent with clinical expectations. Age, BMI, glucose level, hypertension, heart disease, and smoking status all represent medically meaningful stroke-risk factors. The final model should be interpreted as a population-level risk stratification aid rather than a diagnostic system.

From a medical perspective, the most important success criterion is sensitivity to actual stroke cases. That is why the project accepts lower precision in exchange for substantially fewer missed stroke cases.

Study Limitations

1. The dataset is relatively small and heavily imbalanced, with only 249 positive stroke cases.
2. The data come from a public secondary source rather than a single clinical institution, which limits external validity.
3. The best model still produces many false positives, so its practical role would be screening support rather than autonomous diagnosis.
4. PCA improves performance slightly but reduces direct feature interpretability.
5. Literature comparisons are imperfect because different studies use different balancing strategies, split protocols, and evaluation criteria.

References

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